



Compute

Roofline Analysis

root · L1–L4



GPU Compute

Architecture

Networking

Collective Comm.

AllReduce, Ring, Tree

cross-area

Parallelism (6 topics)

Data Parallelism

root · L2–L5

Model / Tensor Parallelism

Pipeline Parallelism

3D Parallelism

requires all three · L5–L6+

Gradient Sync.

requires Data Par. + Coll. Comm.

Scheduling & Resource Mgmt

root (no prereqs)

